Palmeiras Midterm Presentation

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The Problem

- Windows often get dirty
- Washing windows aren't a priority
- Window washing requires going to the outside of the window, which can frequently be difficult to reach or inconvenient if not on ground floor
- Common problem, can be observed anywhere, by anyone who looks through the window

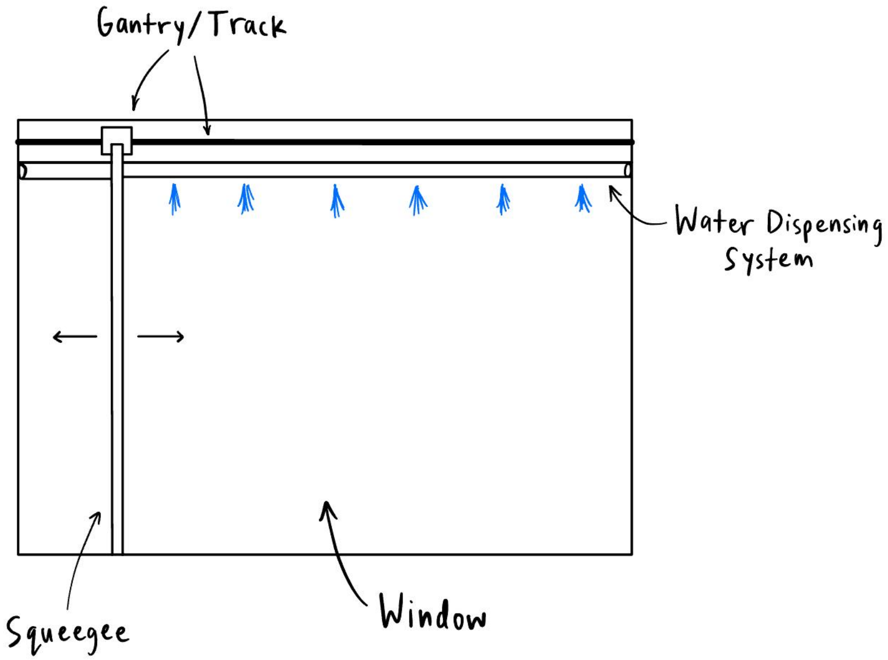
Rec Cen

Home

Ell (Second Floor)

Proposed Solution

- Discrete automatic window cleaning device attachable to any residential window
- Periodically sprays water and cleans window with a squeegee
- Connected to power and water supply so it does not need to be accessed or tampered with often



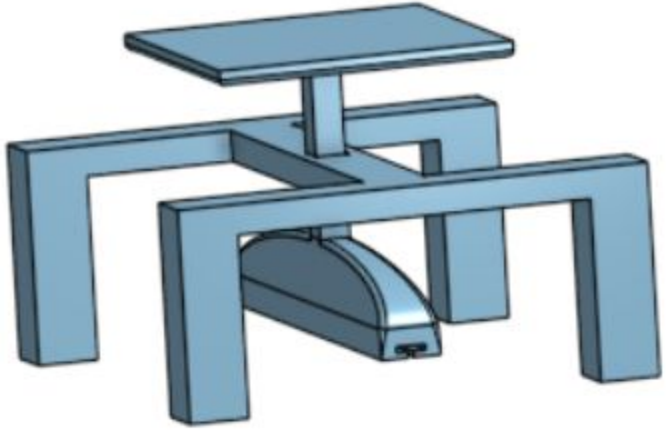
Prototype 1 (Critical Function)

- Test force needed for squeegee/wiper to clean window
- Designed to have weight on the squeegee as it's moved across the window
- Need a squeegee/window wiper and weights to increase the force of wiper on window

$$m_{system} = 576g \longrightarrow F_{squeegee} = m_{system}\, g$$

$$\longrightarrow F_{squeegee} = (0.576kg)(9.81 \tfrac{m}{s^2}) = 5.65N$$

Prototype 1 Calculation (per 4in. of squeegee)



Prototype 1 Initial Design

Prototype 1 (Critical Function)

- Prototype 1 was a success as it quantified the necessary force on the squeegee in order for it to clean the window
- Learned how to attach the squeegee and how much force to apply in order to get ideal clean
- Could have used a different design that minimized friction from the supports

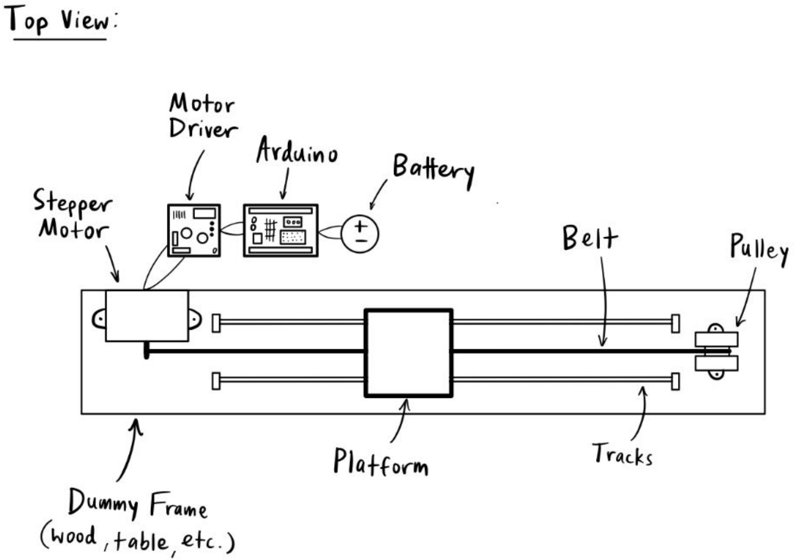
Video Demonstration

Critical Function Data

Weight of System (g)	33	
Additions (g)	Total (g)	Evaluation
122	155	Very nice, not good with sticky
382	415	Good, still leaves some sticky Residue behind
543	576	Cleans most all sticky residue well

Prototype 2

- Must have movement of gantry in the x-direction and be able to wipe a section of window using the squeegee
- Gantry moved using a stepper motor pulley system and controlled by an arduino and driver on a V-slot track



Original Prototype 2 Sketch

Prototype 2

5.65N needed per 4 in. of squeegee (from crit. function)

Say window is 3 ft. tall → need 50.86N on squeegee

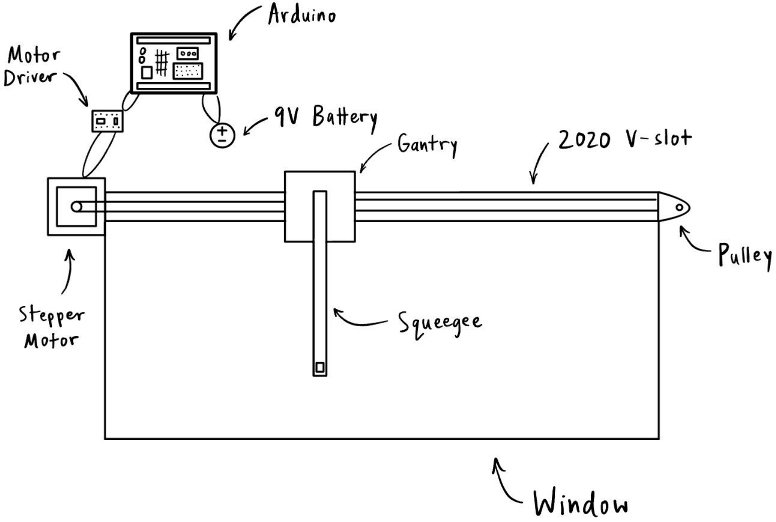
Estimate $\mu_{rubber/water/glass} \approx 0.3$

$$\rightarrow F_f = \mu F_N = 0.3\,(50.86N) = 15.26N$$

$$\tau_{motor} = rF = (0.008m)(15.26N) = 0.122\,N\cdot m$$

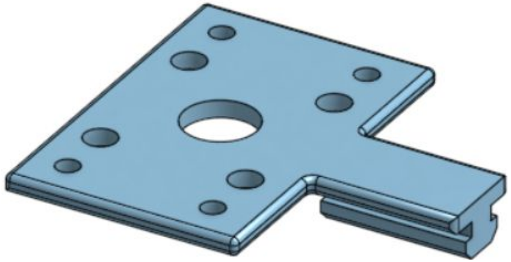
Need 0.122N·m of torque

Motor Torque Calculation

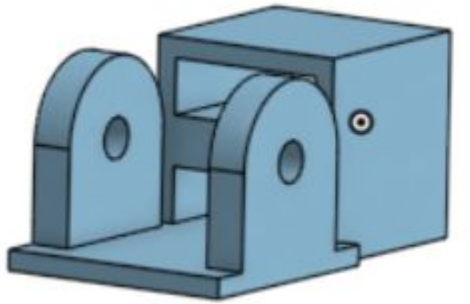


New Prototype 2 Sketch

Prototype 2 - Designs and Components

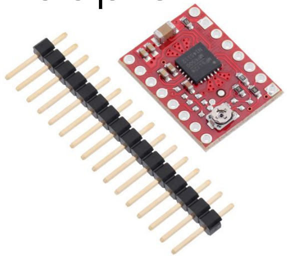


Motor Mounting Adapter



Pulley end adapter

High torque stepper Motor



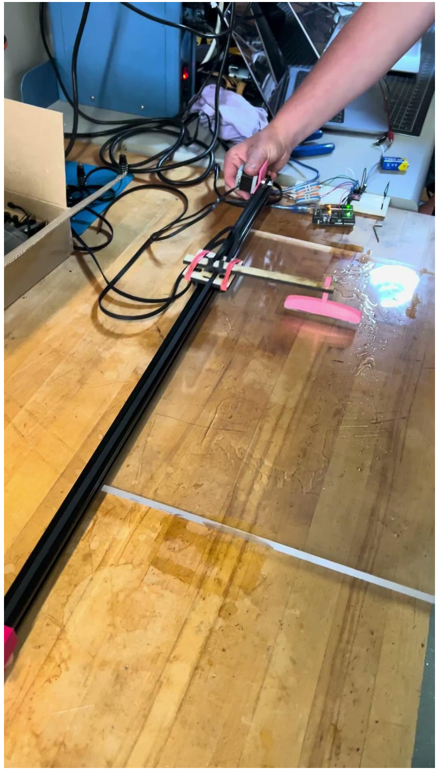
MP6500 Driver

Belt and pulley wheel set

Linear guide on 2020 aluminum beam

Prototype 2

- Prototype 2 was a success!
- This prototype proved that our wiping linear guide design works
- Improvements needed:
  - Add tension and to the belt and cut to right size
  - Program the motor to span the entire V slot with limit switches
- Learned that friction between squeegee and dry test acrylic is high, should wet before running and test on actual window
- Need 12 Volt voltage supply minimum

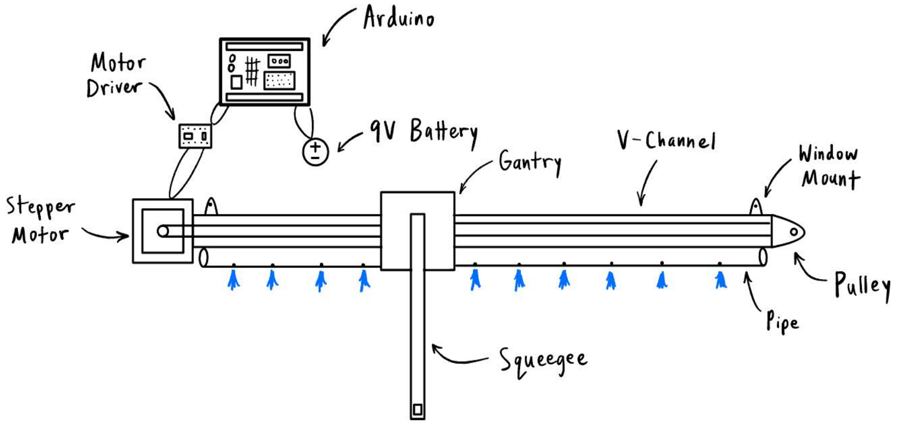


Prototype 2 Video

Prototype 3

- Must have water dispensing system and window mounting figured out. Homing the machine on start up, plug into wall.
- Materials Needed:

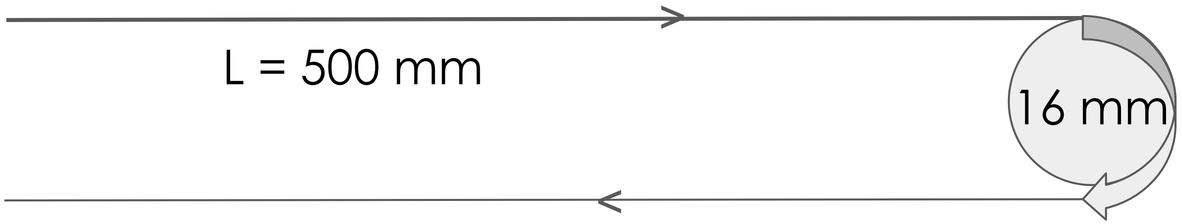
- Piping (Home Depot/Amazon, ~$5, <1 hour)
- Limit Switches (Amazon - $5, 1day)
- 3D printed parts (Makerspace, free, 1 day+)
- Screws, fitting, etc. (Lab, free, no time)
- Test window (Home Depot, $, depends)



Prototype 3 Sketch

Prototype 3 (Calculations)

- For L = 500 mm and D = 16 mm
- $$N\pi D = L$$
- N = Number of rotations = $\frac{L}{(\pi D)} = 9.95$ rotations to cover this distance



Plan for Finishing

Schedule

Task	Week
Water Dispensing System	Week 7/8
Window Mounting	Week 7/8
Force on Squeegee	Week 8
Power Connection	Week 8
Squeegee Attachment	Week 7/8
Electronics Housing	Week 9
Final Clean Up	Week 9

Budget Estimate

Item	Cost
Belts	$15
Gantry	$18
V Channel	$35
Motor	$36
Arduino	$0 (worth $20)
Wipers	$8
Motor Driver	$10
Pipe	~$5
Limit Switch	~$5
Window	N/A

Total Cost = ~$150